

OOP Practice Worksheet (Story Problems)

Q1: Player Health

Ali and Sara are two friends playing an online battle game called BattleZone. Each player has a username visible to others, but their health must remain hidden so no one can cheat. Every new player starts with 100 health. Players can take damage (reduces health) or heal (increases health). Health must never drop below 0 or exceed 100. The game must also track the total number of players created. Task: Design a Player class to model this system. Decide how to handle public/private attributes, enforce health limits, count players, and implement methods for damage and healing.

Q2: Loan Interest

A company offers loans to customers. Each loan has a principal amount and an interest rate. The interest rate must always stay between 1% and 20%; otherwise, the loan becomes invalid. Managers should be able to read the current rate and safely update it. Task: Design a Loan class that safely stores and manages interest rates. Decide how to prevent invalid rates.

Q3: Customer Records

A bank wants to store customer information. Each customer has a unique ID and a name. Employees should be able to add new customers, look up customer details, update a customer's name, and remove customers if they leave. Task: Design a CustomerManager class that supports Create, Read, Update, and Delete operations.

Q4: Loan Counting

A bank wants to count how many loans it has issued. Each time a new loan is created, the counter should increase. Managers want a method that shows the total number of loans issued so far. They also want a quick utility to check whether a proposed interest rate is valid without creating a loan. Task: Design a Loan class that tracks total loans, provides a method to return the count, and a utility method to validate rates.

Q5: Customer Portfolios

Ali and Sara each have multiple bank accounts. Each customer should be able to see their total balance across all accounts. For loans, the bank wants to generate a repayment schedule that shows yearly payments as a list of tuples (Year, Payment). Task: Design classes for customers and loans to manage multiple accounts and repayment schedules.

Q6: Final Comprehensive Story

FinTech Bank is launching a new system. Each BankAccount has an owner (public) and balance (private). Customers can deposit and withdraw. The system must track total accounts created. Each Customer can have multiple accounts and compute their total balance. The BankSystem must manage all customers (CRUD), provide a class method to return total accounts, and a static method to validate transaction amounts. Task: Implement BankAccount, Customer, and BankSystem classes. Create 2 customers (Ali and Sara) with 2 accounts each, perform transactions, and print balances and totals.